ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems

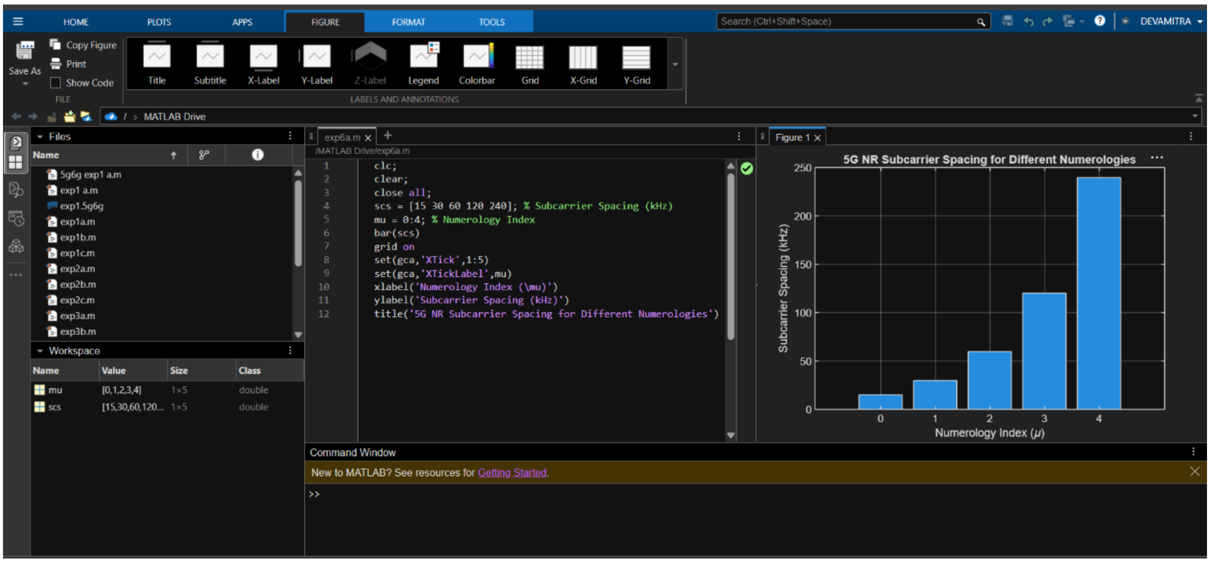
Experiment: 6 - 5G NR Frame Structure Analysis with Different Subcarrier Spacing.

Objective – 1: Analyze Different Subcarrier Spacing Values:

Program:

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Output:

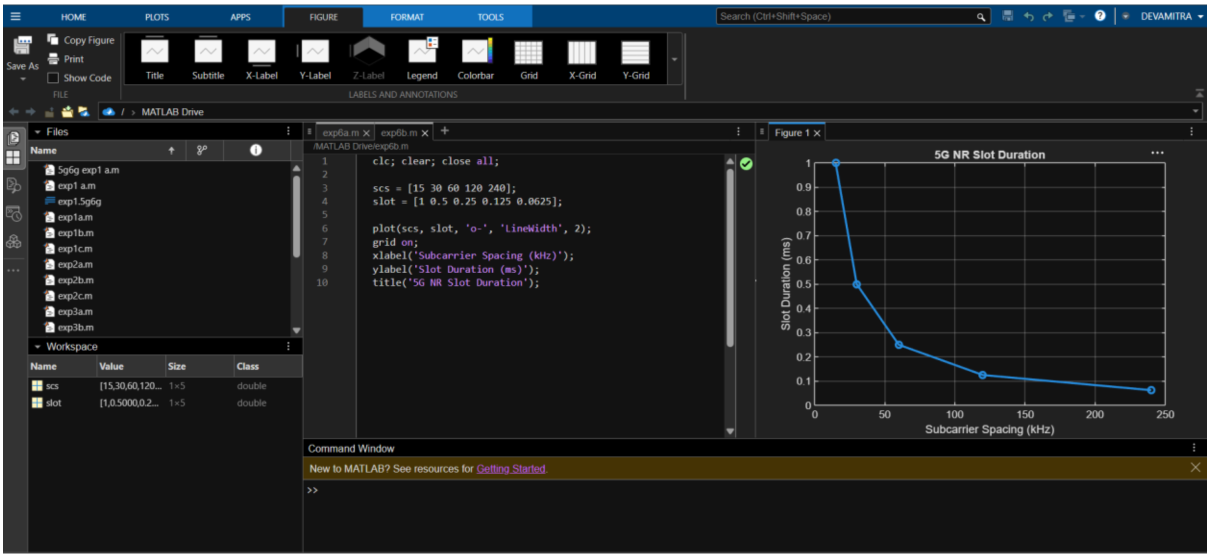


Objective – 2 Compare Slot Duration:

Program:

```
clc; clear; close all;

scs = [15 30 60 120 240];

slot = [1 0.5 0.25 0.125 0.0625];

plot(scs, slot, 'o-', 'LineWidth', 2);

grid on;

xlabel('Subcarrier Spacing (kHz)');

ylabel('Slot Duration (ms)');

title('5G NR Slot Duration');
```

Output:



Objective – 3 Study the Number of OFDM Symbols

Program:

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
```

figure

bar(scs, symbols)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])

Output: